Computational Modeling for Psychology Students

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Contents

1 Identifying Details

PSYCH420 Section 001
Meets: Friday 8:30-11:20
Room: EV3

2 What will we study and how will we study it?

1. What is the difference between a mind and a brain?
2. What is the difference between computational psychology and computational neuroscience?
3. Is the a difference between a computational model of X, and the assertion that X is computational?
4. What is computation?
5. What is cognition?
6. What does it mean to assert that cognition is computation?

2.1 What are Marr's three levels and how do they relate to the question of whether cognition is computational?

David Marr proposed three levels of analysis for understanding cognitive systems:

1. Computational Level: Defines the goal or problem the cognitive system is solving and the logic/strategy for solving it. For cognition, this means specifying what information processing task is being accomplished.

2. Algorithmic Level: Describes the specific representation and algorithm used to implement the computational goal. Here, one details the precise steps and data structures used to achieve the computational objective.

3. Implementation Level: Examines the physical mechanisms that actually realize the algorithm, such as neural structures in the brain.

Regarding computational cognition, Marr's levels suggest cognition can be understood computationally: If cognitive processes can be systematically described at the computational and algorithmic levels, with clear input-output mappings and step-by-step information processing, this supports viewing cognition as fundamentally computational. The implementation level reveals how these computational processes are physically instantiated.

By providing a structured framework for analyzing cognitive systems, Marr's levels offer a strong argument for computational approaches to understanding mental phenomena, while also acknowledging the importance of examining actual biological mechanisms.

Claude$_{\text{AI}}$ *<2024-11-25 Mon>*